

Proposed direction for Question 3: synthetic altitude data and feature correction

High-level approach to cross-altitude transfer in UAV vision

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Executive view

The central hypothesis is that labelled imagery collected at 30 m can be transformed into useful approximations of 80 m imagery, then used to improve a drone vision model at 80 m. The study should test **whether the transformed data improves recognition on genuine 80 m imagery**. It should not assume that one transformation can perfectly recreate everything an 80 m camera would observe.

The strongest first version is deliberately simple. Begin with geometry-informed, anti-aliased scale reduction. Compare it with ordinary scale augmentation and with a mixture of real 30 m and synthetic 80 m training data. Treat blur, noise and haze as optional experimental factors that must be justified by real calibration images. Validate image geometry and appearance, but select the final method using object-detection or segmentation performance on held-out real 80 m data.

This is a study proposal rather than a completed experiment. It defines the direction, comparison groups and evidence needed before a detailed implementation is fixed. No accuracy, runtime or successful-flight result is claimed.

For Question 3, image transformation is a supporting experiment rather than the complete answer. The direct test must still ask whether a single orthogonal transform can align real 80 m features with the 30 m feature space while the vision encoder and the linear probe trained at 30 m remain fixed. The transformed images help determine whether scale and image formation explain the feature shift, and whether a correction learned from synthetic pairs transfers to real 80 m features.

1 Project context

The supplied proposal asks whether altitude changes vision-foundation-model features and, in Question 3, whether a single orthogonal transformation can map features at 80 m onto the 30 m manifold well enough for a linear probe trained only at 30 m to transfer directly to 80 m.

The proposed direction approaches that question at two connected levels. At the image level, labelled 30 m images are transformed to approximate the scale and observable appearance of 80 m captures. These data can be used to retrain or fine-tune a practical detection or segmentation model and test whether explicit altitude simulation improves performance. At the feature level, frozen-encoder representations from real 30 m, synthetic 80 m and real 80 m matched patches are compared. Orthogonal mappings are then fitted on calibration pairs and evaluated on held-out real 80 m objects using the unchanged 30 m linear probe.

The two levels must remain distinguishable in the results. Successful retraining would show that the synthetic data is useful for adaptation, but would not prove that the altitude shift is an orthogonal feature transformation. Successful held-out transfer with a fixed encoder and fixed

probe would address Question 3 directly. If a mapping estimated from synthetic-80/real-30 pairs also corrects real 80 m features, it would provide stronger evidence that the image transformation captures a meaningful part of the real altitude-induced feature shift.

2 Study hierarchy

The study should be understood through the following hierarchy. The order expresses evidential priority rather than a strict timetable: the geometry and synthesis pipeline may need to be implemented before the feature comparisons can be run, but the direct feature-correction result remains the principal answer to Question 3.

Component	Role in the study	What it can establish
Primary study: orthogonal feature correction	Fit one 80-to-30 m mapping on matched calibration objects; evaluate held-out real 80 m features with the encoder and 30 m probe unchanged	Whether the correction proposed in Question 3 transfers beyond the fitted pairs
Supporting mechanism study: synthetic versus real feature shift	Compare real-30, synthetic-80 and real-80 features; test a mapping learned from synthetic-80/real-30 pairs on real 80 m	Whether the image transformation captures a feature displacement that generalizes to the real altitude
Practical experiment: synthetic-data retraining	Retrain or fine-tune a detector or segmenter with real 30 m, synthetic 80 m and mixed training data	Whether the transformed images improve an operational prediction model
Deployment check: independent 80 m evaluation	Test frozen-feature correction offline, then log the selected practical model on the onboard computer and an independent flight	Whether the measured benefit survives held-out data and the intended inference pipeline

This hierarchy prevents a useful detector result from being presented as evidence for an orthogonal feature model. It also prevents a low Procrustes fitting error from being treated as sufficient when the fixed 30 m probe does not improve on held-out real 80 m objects.

3 Proposed research question and objectives

Primary Question 3. Can a single orthogonal transformation map real 80 m features into the 30 m feature space well enough that a linear probe trained only on real 30 m features transfers to held-out real 80 m data, without updating the encoder or the probe?

Supporting image-domain question. Does training with geometry-informed synthetic 80 m imagery derived from labelled 30 m captures improve performance on independent real 80 m imagery beyond ordinary scale augmentation, under matched training conditions?

The proposed approach has seven objectives:

1. verify how object scale changes between 30 m and 80 m for the selected camera and viewing geometry;

2. construct reproducible synthetic training data while preserving labels and object scale at the model input;
3. determine whether additional blur, noise or atmospheric effects are needed;
4. compare the feature displacement caused by synthetic altitude with the displacement observed in real matched 30/80 m pairs;
5. estimate an orthogonal correction only from calibration pairs, then evaluate it on held-out physical objects using a fixed 30 m linear probe;
6. compare feature correction with conventional and altitude-specific retraining strategies; and
7. evaluate the selected practical model offline and during a separate 80 m test flight.

4 Recommended study direction

4.1 Stage 1: establish the geometry

For the same camera, lens, resolution and nadir view over approximately flat ground, ground sample distance (GSD) changes linearly with height. The expected linear object-scale ratio is

$$s = \frac{30}{80} = 0.375, \quad s^2 = 0.140625.$$

An object’s pixel width and height at 80 m should therefore be about 37.5% of their 30 m values, and its pixel area about 14.1%, before other imaging effects. This theoretical ratio should be checked using stable objects or measured ground targets in paired calibration captures. GSD is an appropriate starting model, but it does not account for terrain, object height, oblique views or camera setting changes. [PIX4D’s GSD documentation](#) and [Messmer et al.’s Adaptive Resizer study](#) provide the relevant geometric basis.

A key limitation is field of view. At 80 m, the same camera sees approximately 2.667 times as much ground in each linear direction and 7.11 times as much ground area. Shrinking one 30 m frame cannot reveal the surrounding content. The first implementation should therefore use fixed-size native-pixel tiles from sufficiently broad 30 m coverage or a documented training-only mosaic. It should not describe a small resized image as a complete synthetic 80 m frame.

4.2 Stage 2: create candidate synthetic data

The baseline transformation should reduce the image and its annotations using the verified scale ratio. Compare complete, reproducible resampling implementations such as area reduction, anti-aliased bicubic and anti-aliased Lanczos. Record the software version, output dimensions and anti-alias settings. Official [OpenCV](#) and [scikit-image](#) documentation support area resampling and low-pass filtering for reduction.

The data loader must then be audited. If the shrunken image is automatically enlarged back to a fixed network input size, the intended change in object scale can disappear. The study should inspect the final tensors and labels presented to the model, not only the saved image files.

Start with no extra degradation. Higher altitude does not, by itself, prove that the image must contain more Gaussian blur, sensor noise or haze. Exposure, motion, focus, atmospheric conditions and the camera’s internal processing all matter. Add a residual blur, noise or haze treatment only if real 80 m calibration data shows a consistent mismatch after resampling. Sensor noise should not be treated as a height-only effect; signal-dependent shot and read noise are described by Brooks et al. 2019.

4.3 Stage 3: validate the simulation

Register matched 30 m and 80 m calibration views using stable ground features. A homography may be adequate for a near-planar nadir scene, but one plane cannot simultaneously align the ground, tree crowns and building roofs. The limits of this model should be recorded rather than hidden by repeated warping. The [OpenCV homography tutorial](#) describes this planar assumption.

Validation should be separated into three levels:

Level	Example measures	Purpose
Geometry	Check-point error, object-size ratio, GSD residual	Confirm the intended scale and alignment
Appearance	PSNR, SSIM, edge spread, contrast and noise summaries on valid overlap	Diagnose remaining image differences
Task utility	Detection AP and recall, or segmentation IoU, on real 80 m data	Determine whether the synthetic data improves the actual task

Image similarity alone should not choose the winning transformation. Smoothing can improve some pixel-level scores while removing details needed for detection. Use appearance measures to diagnose the simulator and held-out task performance to select the training approach.

4.4 Stage 4: construct the direct Question 3 feature experiment

Choose one vision encoder for the first experiment and freeze all of its parameters. Define one reproducible patch-extraction and pooling procedure, then extract three feature sets for every matched object: real 30 m features z_{30} , features from the transformed 30 m image z_{s80} , and real 80 m features z_{80} . Use the same image canvas and preprocessing rules across these sets, and verify that the intended difference in native object scale is still present at the encoder input. Resizing every object tightly to one common size could partially remove the altitude variable and should be reported as a separate scale-normalized experiment. Ground-truth regions may be used for this representation experiment, but that makes it an oracle-region analysis rather than a deployable detector.

Train a linear probe only on real 30 m training objects. It remains unchanged for every Question 3 comparison. Use a separate set of physical objects to estimate the feature correction. After centring the paired calibration matrices X_{80} and X_{30} to obtain A and B , estimate the orthogonal Procrustes mapping

$$R^* = \arg \min_{R^T R = I} \|AR - B\|_F^2, \quad A^T B = U \Sigma V^T, \quad R^* = UV^T.$$

Apply the fixed mapping to held-out real 80 m features and pass the corrected features to the unchanged 30 m probe. This is the direct test of Question 3. Standard orthogonal Procrustes provides the least-squares orthogonal solution, but it does not guarantee that altitude effects are actually orthogonal. [Schönemann’s original formulation](#) provides the mathematical basis.

The feature dimension may be much larger than the number of independent calibration objects. A centred calibration set of n objects has rank at most $n - 1$, so a full-dimensional rotation may be underdetermined outside the observed subspace. The first analysis should therefore compare the full representation with a lower-dimensional projection fitted only on calibration or development data. The projection dimension must be selected without using final test objects.

4.5 Stage 5: compare feature correction and image-domain adaptation

The core Question 3 comparisons should be evaluated on the same held-out real 80 m objects:

Arm	Treatment	Interpretation
Q3-A	Fixed 30 m probe evaluated on held-out real 30 m features	Source-domain reference
Q3-B	The same probe evaluated directly on real 80 m features	Uncorrected cross-altitude transfer
Q3-C	Real 80 m features mapped by Procrustes fitted on separate real 30/80 m calibration pairs	Direct, target-calibrated test of Question 3
Q3-D	Mapping fitted between synthetic 80 m and real 30 m calibration features, then applied unchanged to real 80 m features	Tests whether image synthesis captures a transferable feature shift
Q3-E	Mean correction or a regularized linear mapping fitted on the same calibration pairs	Tests whether orthogonality is the useful constraint

Report both probe performance and feature alignment. Higher alignment similarity without better held-out classification is not sufficient. Q3-C uses real 80 m information to estimate the mapping and must be described as target-calibrated. Q3-D is the most direct bridge from image synthesis to the original research question: it asks whether a correction derived without real 80 m feature pairs generalizes to the real target altitude.

A separate practical experiment may retrain or fine-tune a detector or segmentation model. Use four initial arms: real 30 m only; real 30 m with ordinary scale augmentation; synthetic 80 m only; and a mixture of real 30 m and synthetic 80 m. Add calibrated degradation only after the image comparison shows a residual mismatch. This experiment establishes whether the transformed data is useful for adaptation, while Q3-A through Q3-E establish whether frozen-feature correction is possible.

The direct published precedent is [Rüter et al. 2024](#), which found that transformed lower-altitude imagery improved high-altitude human detection in its particular grass-scene dataset. Its use of real target-altitude statistics and its weaker transfer to a different dataset mean that it supports feasibility, not a prediction of this project’s result.

4.6 Stage 6: evaluate on independent real 80 m data

Split the dataset by physical object, geographic area and flight session before creating synthetic derivatives. Adjacent video frames, all altitudes of one physical object and every transformed version of a source image must stay in the same split. Keep the real 80 m pairs used to estimate or select a mapping separate from model validation and final testing.

First evaluate the feature mappings and practical models offline on the same recorded 80 m imagery. Then deploy the selected practical model to the drone computer and verify the complete input and inference pipeline. A subsequent independent 80 m flight should run the model in logging or shadow mode, recording predictions, missed frames, latency and hardware conditions for later annotation and analysis. This evaluates perception and onboard integration without treating one successful flight as proof of autonomous-flight safety.

5 Data and evaluation plan

Start with the 30/80 m nadir condition before expanding to 15, 50 and 120 m or to the 45-degree view. This isolates the endpoints named in Question 3 and reduces geometric variation during method development. Use persistent objects with reliable cross-altitude identity for feature analysis. If a practical detector is also evaluated, use a small set of operationally meaningful classes and include background-only images to measure false detections. Record altitude above local ground, camera intrinsics and recording mode, gimbal angle, exposure, ISO, time, light, wind and flight speed.

The feature experiment should report linear-probe accuracy and macro-F1 on held-out object identities, together with calibration residual, cosine similarity and class-wise effects before and after mapping. The principal Question 3 contrast is Q3-C versus Q3-B. Q3-D versus Q3-B tests whether the synthetic transformation produces a mapping that transfers to genuine 80 m data. Results on real 30 m data provide the reference for how much of the source performance is recovered.

For the auxiliary detection experiment, report mean average precision across IoU thresholds 0.50–0.95, AP50, class-level AP, recall, false positives and results by native object size. Keep model architecture, initialization, input pipeline and optimization budget matched across training arms. Repeat stochastic training with paired seeds and report dispersion.

Select all mappings, projections, transformation parameters and checkpoints using training and development groups only. Treat flights, scenes or physical objects as the independent units for uncertainty; large numbers of neighbouring frames are repeated observations. Predefine the principal contrast and a minimum useful improvement before opening the final test results. If the method changes after test inspection, the revised claim requires a new held-out evaluation.

6 Expected contribution and limitations

The study can make a useful contribution even if the synthetic method does not improve the practical model. It can quantify the 30/80 feature displacement, test whether that displacement is sufficiently structured for post-hoc orthogonal correction, and reveal whether image scale, lost resolution, missing context or camera processing explains the failure. A result in which a simple correction matches a more flexible mapping is also informative.

The possible outcomes must be interpreted carefully. If retraining on synthetic 80 m images improves detection but Procrustes does not improve the fixed 30 m probe, the transformation is useful augmentation but does not support the orthogonal-correction hypothesis. If Q3-C succeeds but Q3-D fails, a post-hoc correction may be possible, but real 80 m calibration is still required and the image simulator has not captured the real feature shift. If Q3-D succeeds on held-out real 80 m objects, the synthetic transformation provides evidence for a transferable feature-space correction. If neither succeeds, the result still identifies limits of the single-transform assumption.

The principal risks are missing ground context, irreversible loss of fine detail, non-planar geometry, different camera processing between flights, high-dimensional feature mappings fitted from too few independent objects, and information leakage from repeated scenes. An orthogonal mapping preserves feature-space distances and cannot reconstruct visual evidence that was never captured. The initial sample target in the original proposal should be treated as a planning estimate rather than a statistically proven sample size. Appropriate university, site, privacy and aviation arrangements are also required before collecting imagery over or near people.

The next detailed report can convert this direction into a complete protocol: exact camera equations, file and annotation formats, transformation code, registration procedure, encoder and feature-layer choice, pooling method, Procrustes implementation, split manifest, metric implementation and pre-registered analysis plan.

Selected references

1. Rüter, J., Maienschein, T., Schirmer, S., Schopferer, S., and Torens, C. (2024). *Filling the Gaps: Using Synthetic Low-Altitude Aerial Images to Increase Operational Design Domain Coverage*. *Sensors* 24(4), 1144. [Publisher full text](#).
2. Messmer, M., Kiefer, B., and Zell, A. (2022). *Gaining Scale Invariance in UAV Bird’s Eye View Object Detection by Adaptive Resizing*. ICPR 2022; arXiv:2101.12694. [Paper](#).
3. Kiefer, B., Ott, D., and Zell, A. (2021). *Leveraging Synthetic Data in Object Detection on Unmanned Aerial Vehicles*. arXiv:2112.12252. [Full text](#).
4. PIX4D. *Ground sampling distance in photogrammetry*. [Official documentation](#).
5. OpenCV contributors (version 4.13.0). *Geometric Image Transformations*. [Official documentation](#).
6. OpenCV contributors (version 4.13.0). *Basic concepts of the homography explained with code*. [Official documentation](#).
7. Brooks, T., Mildenhall, B., Xue, T., Chen, J., Sharlet, D., and Barron, J. T. (2019). *Unprocessing Images for Learned Raw Denoising*. CVPR 2019. [Paper](#).
8. Schönemann, P. H. (1966). *A generalized solution of the orthogonal Procrustes problem*. *Psychometrika* 31, 1–10. [Original paper scan](#).
9. The supplied one-page proposal, *Altitude as a Controlled Variable: Probing Scale Subspaces in Vision Foundation Models*, and the student’s five-step plan provide the project context. They are not treated as evidence that the proposed method already works.